

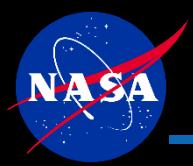
Space-Based Solar Power: An Enabler for Expanded Lunar Surface Exploration and Mobility Operations

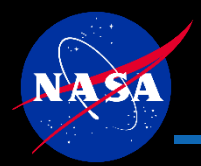
ASTRA-Con 2024
September 19th

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SACD
SYSTEMS ANALYSIS AND CONCEPTS DIRECTORATE





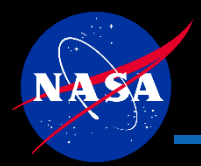
Agenda



- Overview – NASA returning to the Moon
- Challenges on the Lunar Surface
- SBSP for Power Augmentation
- OPAL Study Overview
- Study Approach
- Reference Missions
- Power Classes
- Technology Assessment
- Design and Analysis Cycles
- Forward Work



- NASA is planning to return to the Moon to:
 - » **Discover** natural resources, such as oxygen (from ice or regolith), water, or iron
 - » **Test** human exploration capabilities for Mars
 - » **Learn** about the undiscovered treasures of the solar system



Exploration Challenges on the Moon



- Lunar surface exploration missions will require:
 - Uncrewed and crewed mobility systems for extended exploration on the lunar surface
- Although, there are challenges that the Moon presents that make exploration difficult
 - Not enough sunlight = power restricted
 - Permanently Shadowed Regions (PSRs)
 - The Lunar night
 - Two-week period where the sun will not reach the surface
 - Extreme temperatures
- These constraints make it difficult to explore the lunar surface and require vehicles to be self-sufficient (batteries or photovoltaic (PV) panels)



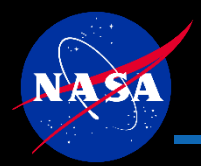
But what if there were an alternative....

An alternative that would enable:

- Constant power
 - Longer traverse capabilities
 - Exploration in previously restricted regions
 - Reduction of surface vehicle mass
 - Search and rescue
- Space-based solar power (SBSP) is a power augmenting alternative that could help to address some of challenges on the Moon and provide power to assets operating on the lunar surface
 - Power-restricted zones
 - PSRs
 - Through the Lunar night



Space based solar-power beaming can be a critical technology to provide power to the lunar surface for assets to survive the lunar night and travel to non-ideal locations for science discoveries and exploration



Study Overview



The goal of the Orbital Power-beaming Assets for Lunar Applications (**OPAL**) Project is to explore the concept of SBSP as a means for **external power augmentation on the lunar surface**. Over the past few months, we have:

- Developed reference missions that reflect the power-demands and operational constraints of planned lunar surface assets
- Conducted a technology assessment of key SBSP spacecraft subsystems
- Utilized parametric design capabilities to estimate the size and mass of SBSP vehicle for applications at different scales



The results of this study will yield a SBSP system design capability for future mission considerations

- The OPAL study is unique in that it focuses on:
 - Tunable system
 - The ability to tune the power density through movable optical components
 - Providing power densities to multiple users
 - Provide various power densities to multiple end users based on their power requirements; matching the irradiance of the sun and beaming to nearly the exact same area as the solar array on the lunar assets
 - Optical power beaming
 - Attractive due to the **smaller beam size** than microwave
 - No atmosphere
 - Unlike Earth, the Moon does not have an atmosphere or cloud cover that could scatter the optical power beam
- Limitations:
 - Power on the SBSP vehicle will be limited by:
 - Power collected by solar arrays
 - Power available on batteries
 - Access times based on orbital parameters





OPAL Approach



- Identifying examples of surface power needs on the Lunar surface
- Ex. LTV, PR, CLPS, ISRU, SH, etc.

Client Identification



- Identifying specific examples of ongoing, planned, or proposed future missions
- Identify specific example power needs and conops
- Identify receiver types and mission limitations

Client Assessment



- Define power classes based on...
- Current and future predicted power needs on the Lunar surface
- Limitations of surface technology

Power Class Definition



- Identify SBSP technology limitations
- Perform a frequency down-selection
- Assess the potential conops, limitations, and sizing possibilities for SBSP

SBSP Technology Assessment



- Generate SBSP parametric design for the top of each power class – preliminary sizing and spacecraft mock-up
- Generate preliminary set of conops for each design

Design & Analysis Cycles



- Review all materials up until this point for the SBSP designs
- Validate whether designs are viable in the near-term, mid-term, far-term, or if it cannot be determined at this time

Design Validation



- Trace client needs back to the SBSP designs
- Consider how conops may change for each client
- Document how SBSP can augment/improve current mission

Client Re-evaluation



Forward Work

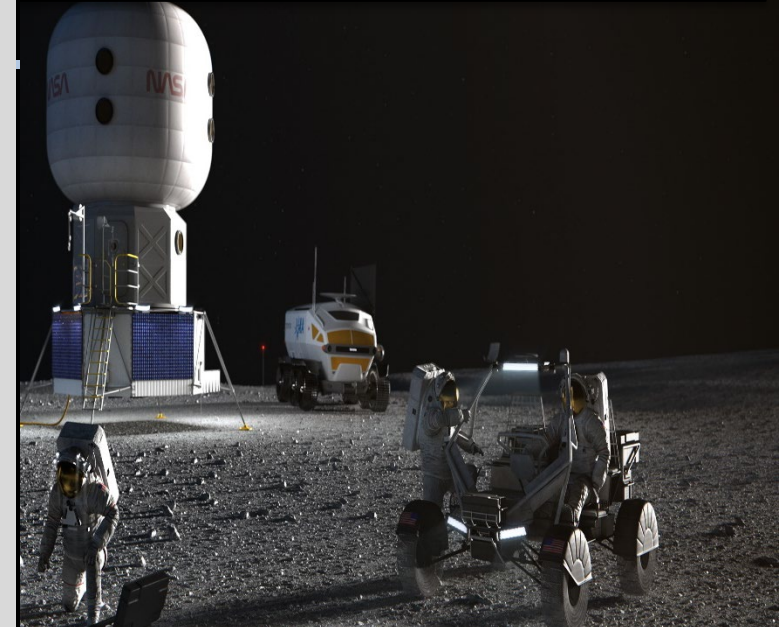
Exploration Inside and Through PSR & Other
Extreme Regions



Search, Rescue, and Contingency Ops



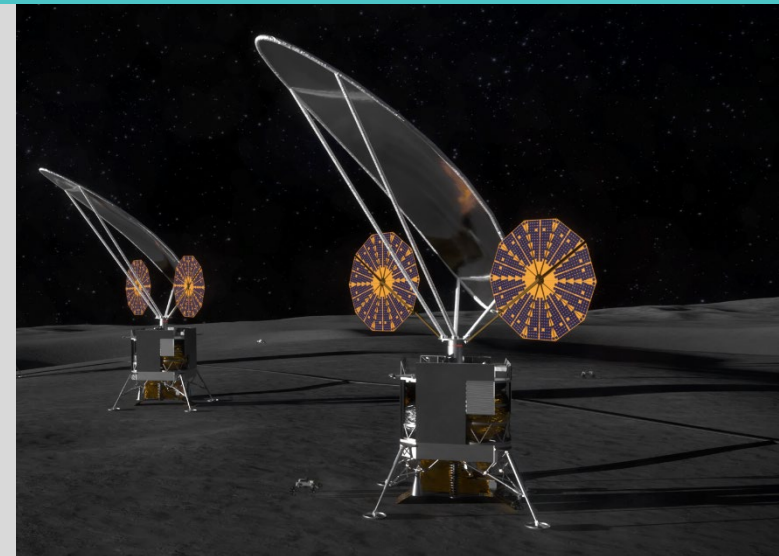
Relocation of Assets from Region-to-Region



Long-Range Rover Traverses

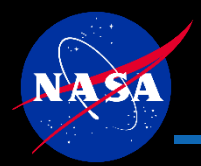


Extended Surface Habitation
(Polar & Non-Polar Regions)



Industrial-Scale In-Situ Resource and Utilization
(ISRU), Expanded Habitation & Commercial Activities

Potential Reference Missions for Lunar SBSP Seedling Study



Reference Missions



- The team defined reference missions in order to determine the amount of power generation the SBSP vehicle will need to produce to service various clients on the lunar surface
- Power classes were defined based on the current and future predicted power needs from rovers and landers on the lunar surface

Class I

Based on current CLPS lander designs

- Astrobotic Peregrine
- IM-1 (NOVA-C)
- IM-2
- IM3

Power Needs: **90 – 200 W**

Average Operating Power: **100 W**

Assumed Solar Array Area: **1 – 4 m²**

Class II

Based on prospecting rover designs

- VIPER
- LTV Survival Mode

Power Needs: **300 – 450 W**

Average Operating Power: **500 W**

Assumed Solar Array Area: **~3 m²**

Class III

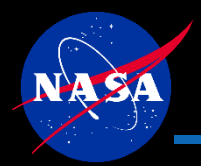
Based on large prospecting rover designs

- LTV Crewed Mode
- PR Survival Mode
- PR Crewed Mode

Power Needs: **1.75 – 8 kW**

Average Operating Power: **2 kW**

Assumed Solar Array Area: **~3 m²**



Technology Assessment Overview

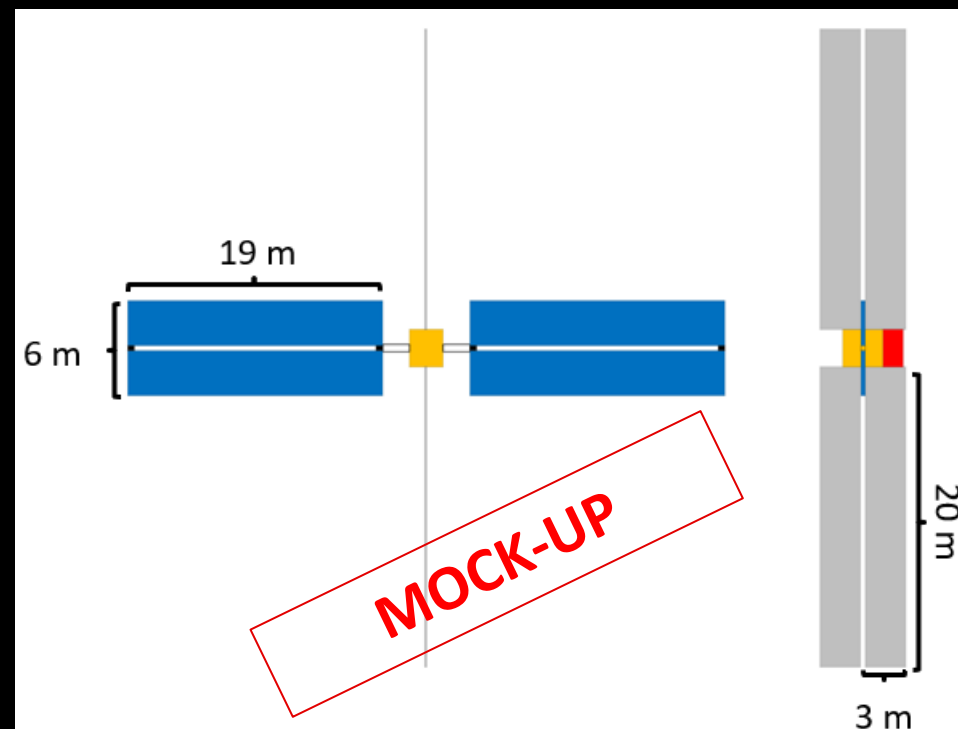


- To understand the trade space and technologies available for SBSP applications, the team performed a technology assessment
- Through this technology assessment, the team wanted to understand the various technologies and how they can be used for vehicle sizing and analysis
- Technologies assessed:
 - Solar cells and efficiencies
 - Solar arrays
 - Vehicle structure (bus size, truss)
 - Laser
 - Batteries
 - Client receivers

DC to Optical Power Conversion Efficiency	25%	30%	35%	40%	45%
Solar Cell Efficiency	19.5%	23.5%	29.5%	30%	33%
PV Array Specific Mass (kg/m ²)	0.10	0.50	1.0	3.0	5.0
Battery Specific Capacity (Whr/kg)	160	180	200	220	240

Preliminary OPAL Spacecraft Trade Space

- Design and Analysis Cycles (DACs) were conducted to determine estimates for the SBSP vehicle size and mass:
 - Team used a NASA internal parametric design tool to determine key sizing variables and parameters
- Key Findings:
 - **Primary drivers** for sizing – Laser Suite (Electronics, Optics & Support Structure), TCS (Acquisition & Rejection), and Electric Power System
 - TCS (Rejection) comprises ~**43%** of spacecraft dry mass for all classes
 - One SBSP vehicle is capable to be packaged in a single Commercial Launch Vehicle fairing (4.6 m)



SBSP Class 1 Spacecraft



WHERE ARE WE NOW? (CDF)

Study Goals

- Understand the performance of a range of space-based solar power (SBSP) concepts for lunar applications.
- Identify and explore the impact of opportunities for improving State of the Art.
- Show benefit of SBSP to augment clients' ability to survive the lunar night, operate in darkness, and expand surface operations
- Develop capabilities to assess mission concepts across the technology trade space for orbital-to-lunar surface power beaming

Completed – Preliminary Results

- Parametric modeling capability to estimate the size and mass of the SBSP spacecraft for all power classes
- Reference missions for SBSP client and developed conops
- Technology assessment of the state-of-the-art of key SBSP spacecraft subsystems

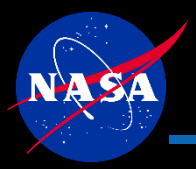
In-Progress

- Finalize Design and Analysis Cycles (DAC)
- Prepare final products for delivery – 9/2024

WHAT'S NEXT? (LaRC FY25 CIF/IRAD)

CIF/IRAD 10/2024 – 9/2025:

- Increase the fidelity of optical SBSP concept modeling capability
- Develop more detailed concept of operations for SBSP spacecraft constellation beaming to multiple surface assets
- Develop top-level tunable laser system requirements driven by high level Concept of Operations



Questions?

